

How to stop

Charles Yang



ON LANGUAGE

The Suffix That Tells Us to
Ruthlessly Optimize Everything

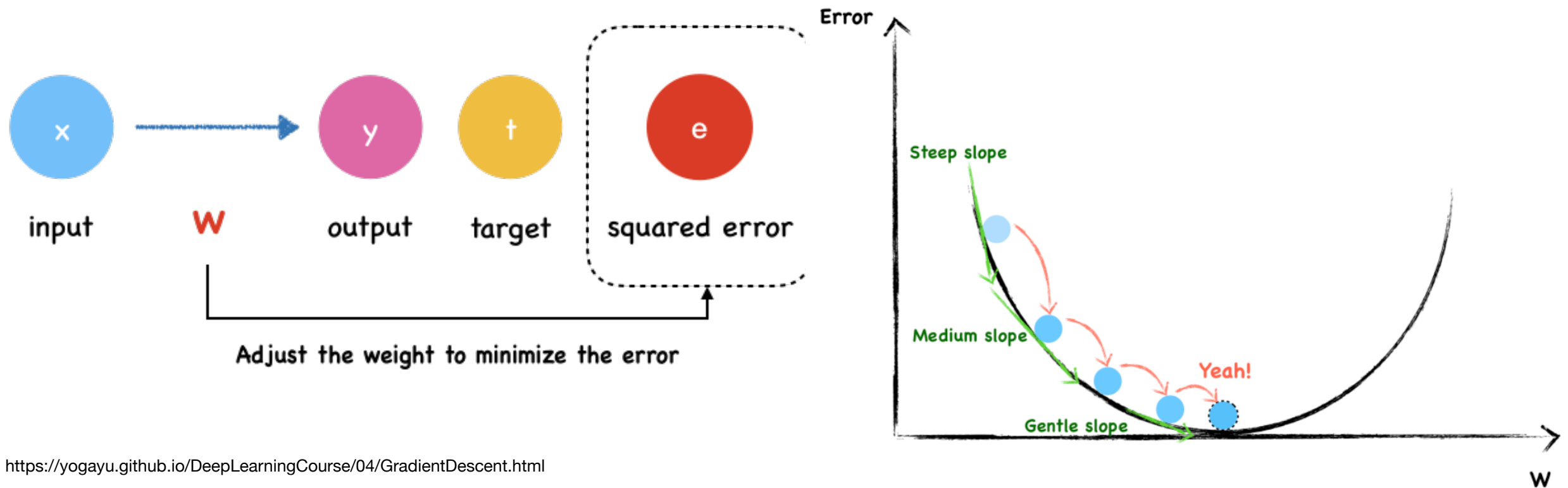
New York Times (Oct 11, 2025)

Reds and Blues

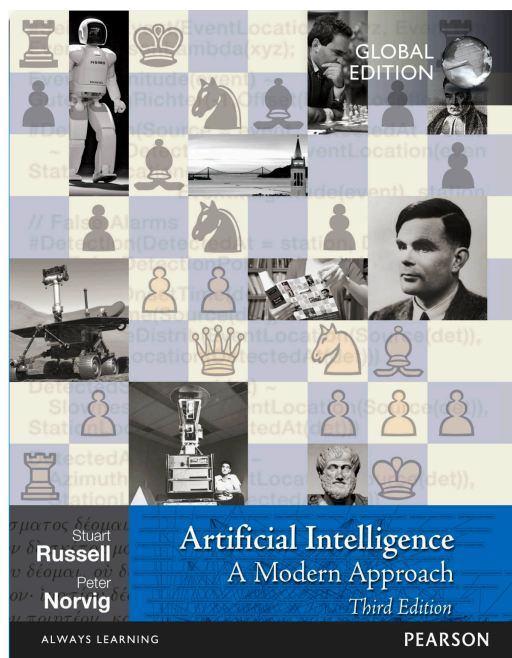
[B R R R B B B R R R B R] [B B R R R B B R R R R B]

- You saw two sequences of 12 colors: 7 reds and 5 blues, randomly mixed
 - No one can retain a sequence of 24 but will have the memory of the most recent colors
- Maximizer: The best option is always Red (100%)
- Luce sampling: Approximating probabilities Red (58%)

Why Maxxing?



<https://yogayu.github.io/DeepLearningCourse/04/GradientDescent.html>



- In this book, we adopt the view that intelligence is concerned mainly with **rational action**. Ideally, **an intelligent agent** takes the best possible action in a situation. We study the problem of building agents that are intelligent in this sense.

The beginning

- The study of the mind dates back to antiquity but why did the field of cognitive science emerge in the 1950s?
- All sciences require a formal language: some are borrowed (evolutionary biology) but some are invented (physics)
- The formal language of cognitive science: The theory of **computation**
 - It's no longer about the description of the data but the mechanisms that generate the data.

Another experiment

dax

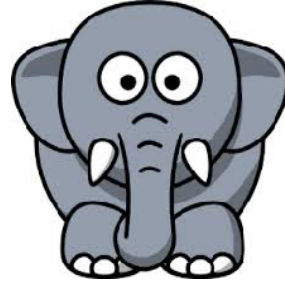


- Each time, you will see two animals on the screen and hear/see a word (**dax**) that refers to one of them
 - Try to figure out which word refers to which animal
- The 'experiment' will take only 30 seconds and you will be given a multiple choice test afterwards
- Based on joint work with Lila Gleitman and John Trueswell

wug



koba



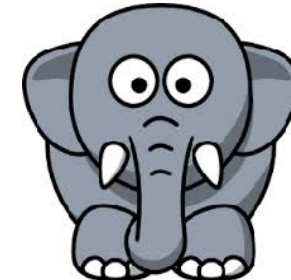
koba



wug



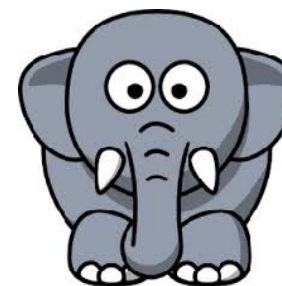
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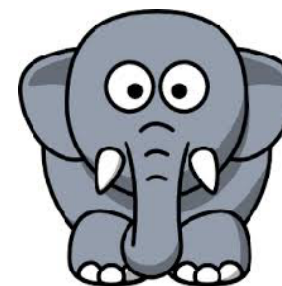
wug



Find wug

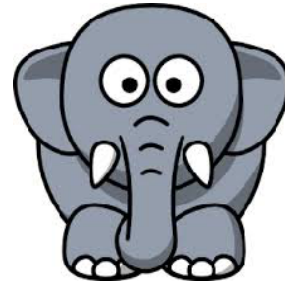


Find koba

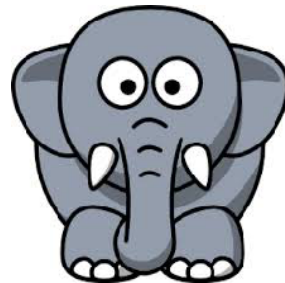


Mipen: One word, Three animals

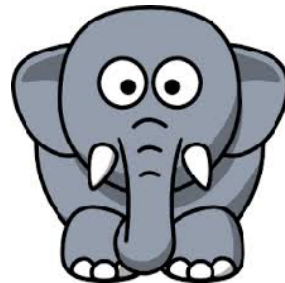
“mipen”



“mipen”



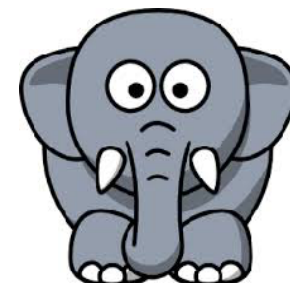
“mipen”



“mipen”



“mipen”



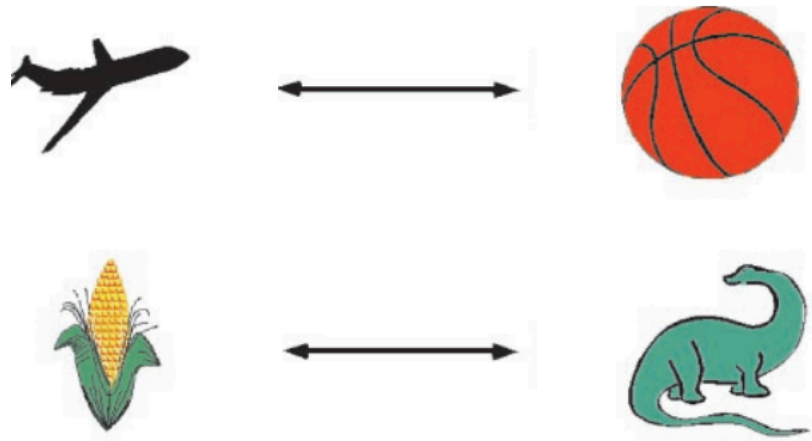
Find mipen!



Introspection

- What do you think you were doing to figure out these word meanings?

Paired associate learning

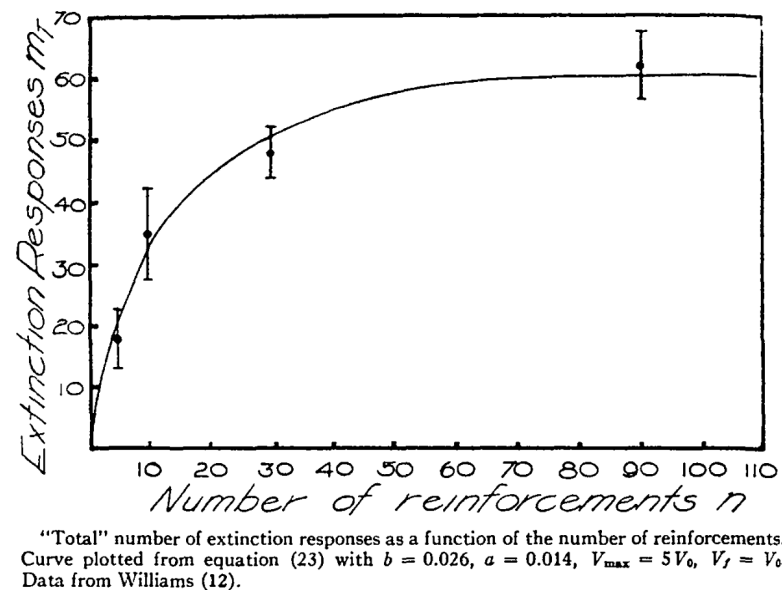


11 $\iff$ A

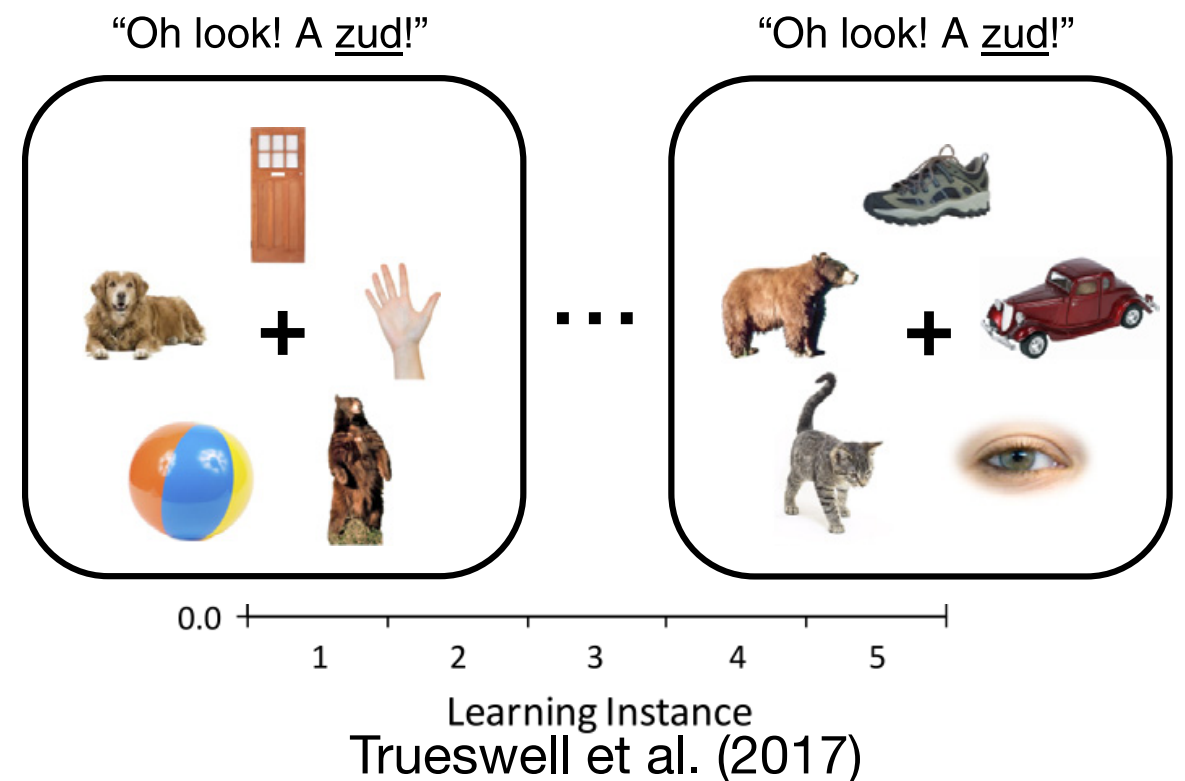
8 $\iff$ C

15 $\iff$ Z

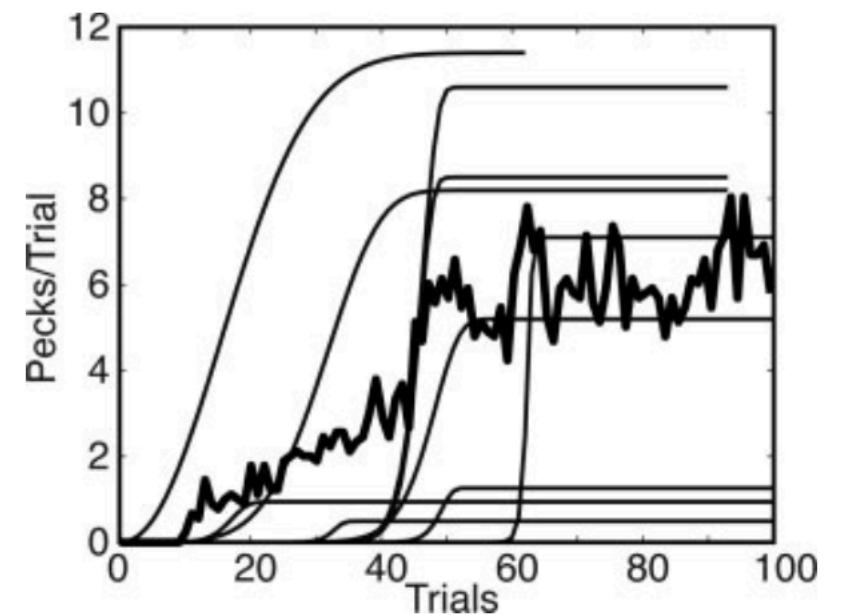
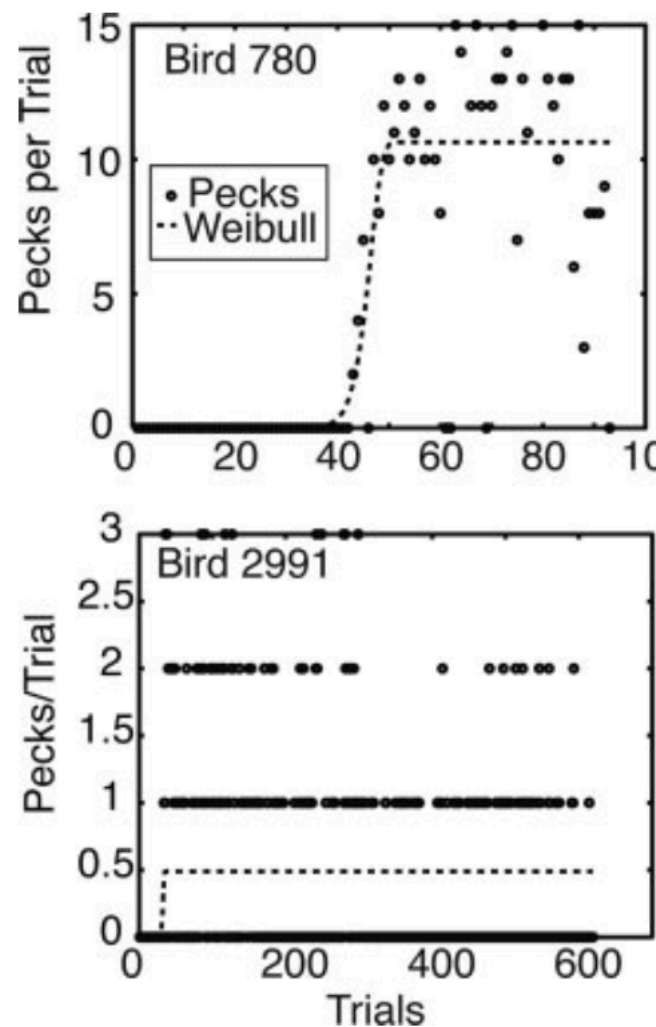
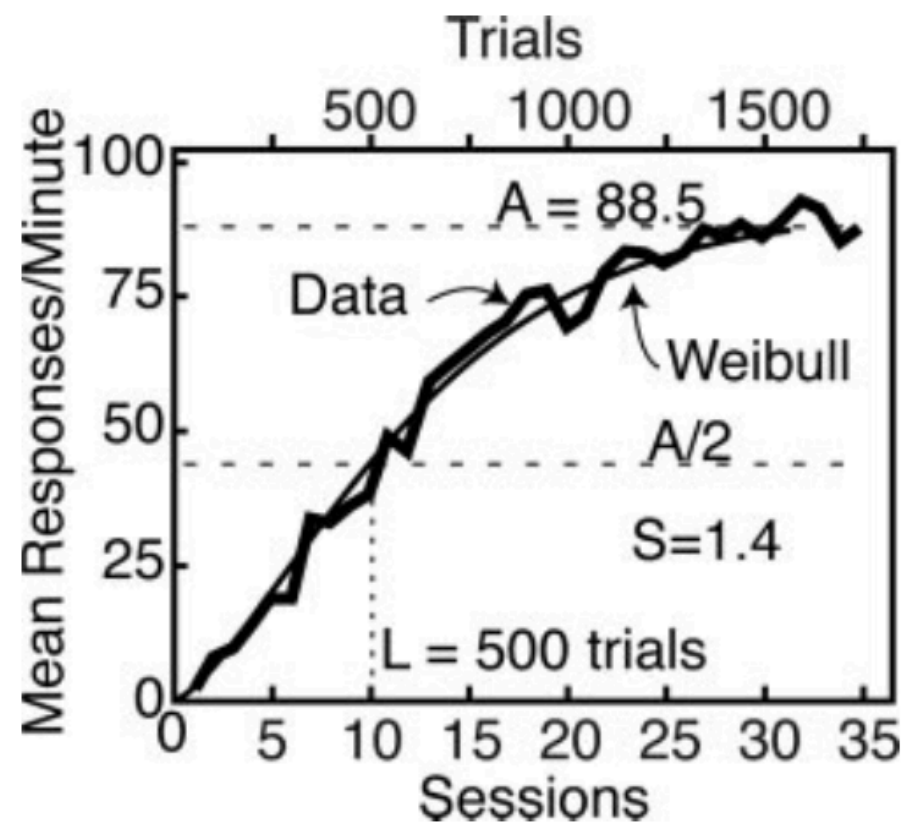
- Extensively studied in the first half of the 20th century: A smooth learning curve is typical



Bush & Mosteller (1951)



Fixing probabilities?



Gallistel et al. (2004)

		know	don't know
		C_1	C_0
know	C_1	1	0
don't know	C_0	c	$1 - c.$

Estes (1951)

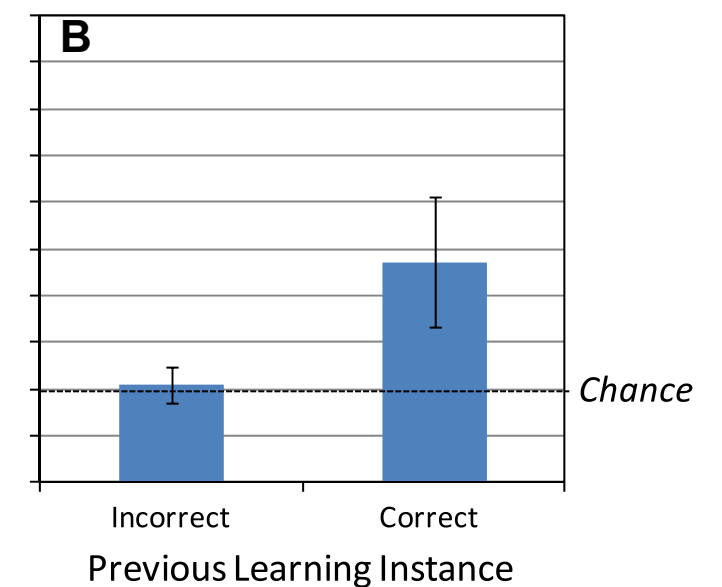
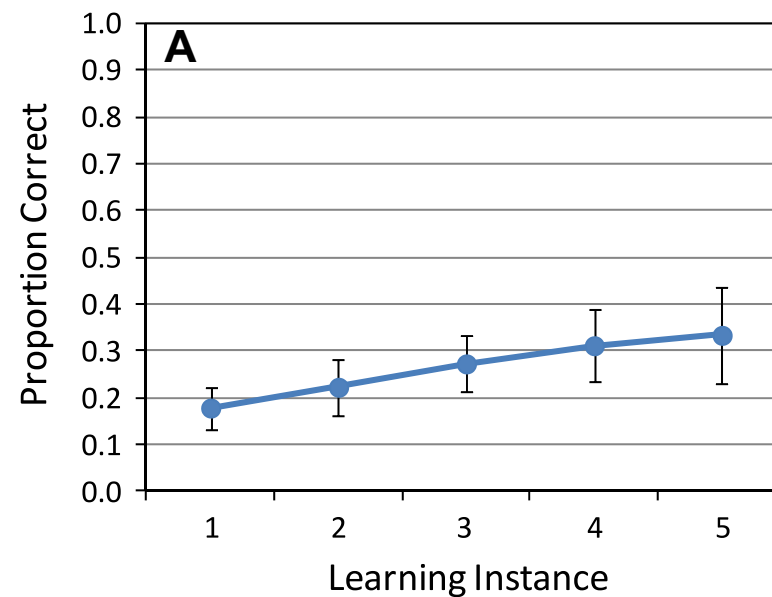
Fixation of belief



Trueswell et al. (2017)

		know	don't know
		C_1	C_0
know	C_1	1	0
don't know	C_0	c	$1 - c.$

Estes (1951)



Back to Red and Blue

- Almost all decisions have to be made with imperfect, incomplete, and inconsistent data
- Nevertheless, we often commit to a decision and act accordingly.

(Bounded) Rationality

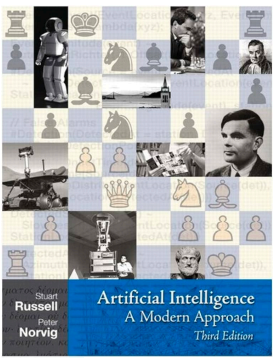


A BEHAVIORAL MODEL OF RATIONAL CHOICE

By HERBERT A. SIMON*

Traditional economic theory postulates an “economic man,” who, in the course of being “economic” is also “rational.” This man

Broadly stated, the task is to replace the global rationality of economic man with a kind of rational behavior that is compatible with the access to information and the computational capacities that are actually possessed by organisms, including man, in the kinds of environments in which such organisms exist.



27.3 ARE WE GOING IN THE RIGHT DIRECTION?

In Chapter 1, we said that our goal was to build agents that *act rationally*. However, we also said that

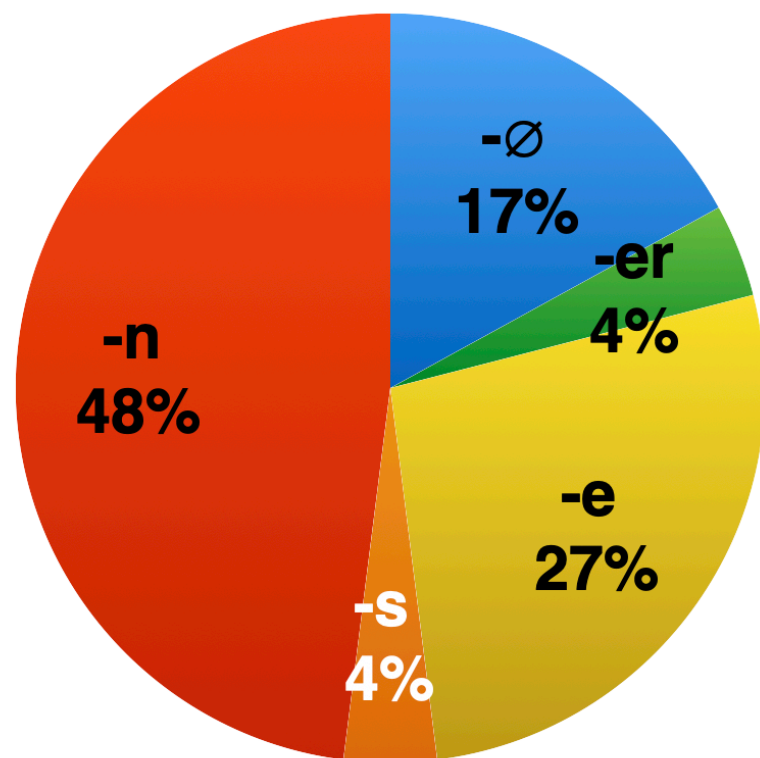
... achieving perfect rationality—always doing the right thing—is not feasible in complicated environments. The computational demands are just too high. For most of the book, however, we will adopt the working hypothesis that perfect rationality is a good starting point for analysis.

The capacity of the human mind for formulating and solving complex problems is very small compared with the size of the problems whose solution is required for objectively rational behavior in the real world—or even for a reasonable approximation to such objective rationality.

He suggested that bounded rationality works primarily by **satisficing**—that is, deliberating only long enough to come up with an answer that is “good enough.” Simon won the Nobel Prize in economics for this work and has written about it in depth (Simon, 1982). It appears to be a useful model of human behaviors in many cases. It is not a formal specification for intelligent agents, however, because the definition of “good enough” is not given by the theory. Furthermore, satisficing seems to be just one of a large range of methods used to cope with bounded resources.

Language: The Paradigm Case

- “All grammars leak” (Sapir 1928)
- English past tense rule is “add -ed” but there are some 100 irregulars that do not.



Autos, Parks, Pizzas, ..., iPhones

- End in “e”: add -n.
- End in “el/er/en”: do nothing.
- Is masculine: add -e.
- Is feminine: add -en.
- ...
- ...
- Altogether about 15% of nouns require rote memorization.

The best is not good enough

I **stride** down the street.
 You **strode** down the street.
 They have **???** down the street.

(6) Third-conjugation mid vowel stem change patterns:

sumergir 'to submerge' (no change):

pres. indic.	sumerjo	sumerges	sumerge	sumergimos	sumergís	sumergen
pres. subj.	sumerja	sumerjas	sumerja	sumerjamos	sumerjáis	sumerjan

discernir 'to distinguish' (diphthongizing):

pres. indic.	discierno	disciernes	discierne	discernimos	discernís	disciernen
pres. subj.	discierna	disciernas	discierna	discernamos	discernáis	disciernan

desvestir 'to undress' (lowering):

pres. indic.	desvisto	desvistes	desviste	desvestimos	desvestís	desvisten
pres. subj.	desvista	desvistas	desvista	desvistamos	desvistáis	desvistan

agredir 'to attack' (defective):

pres. indic.	*	*	*	agredimos	agredís	*
pres. subj.	*	*	*	*	*	*

do+not = don't

are + not = aren't

is + not = isn't

does + not = doesn't

...

am + not ≠ **amn't**

may + not ≠ **mayn't**

The Bronx
 The Hague

***The Berlin**

***The Chicago**

***The Montreal**

***lažu** 'I climb'

***pobežu** (or ***pobeždu**) 'I conquer'

***deržu** 'I talk rudely'

***muču** 'I stir up'

***erunžu** 'I behave foolishly'

a. je fris, tu fris, il frit, **nous ???, vous ???, ils ???**
 I fry, you.sg fry, he fries, **we ???, you.pl ???, they ???**

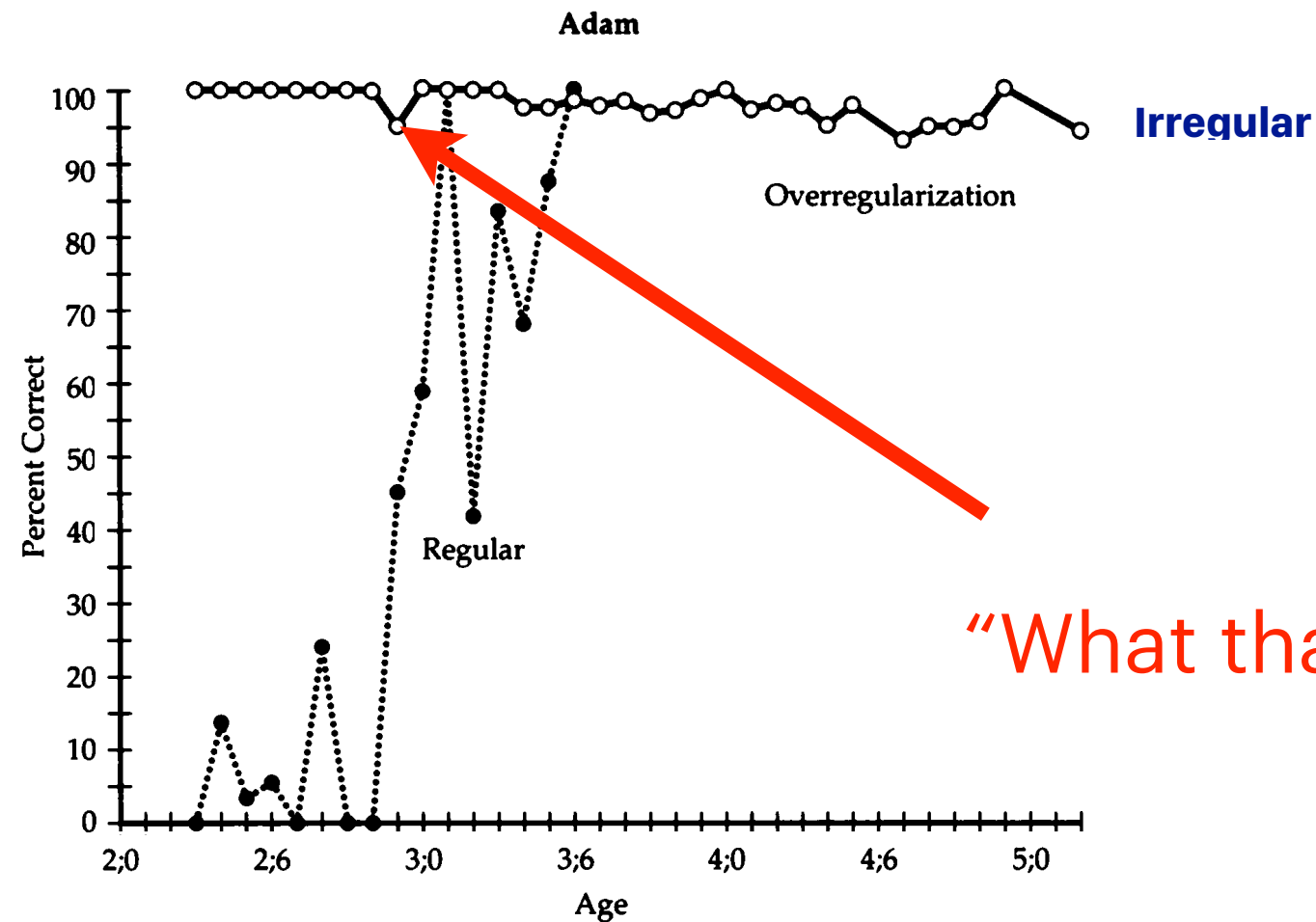
b. je clos, tu clos, il clôt, **nous ???, vous ???, ils closent**
 I close, you.sg close, he closes, **we ???, you.pl ???, they close**

What did John see that Bill ate *t* ?

*What did John complain that Bill ate *t* ?

*What did John quip that Bill ate *t* ?

Threshold in Language Learning



“What that feeled like”?

Non-monotonic learning is common in child language.

"I died him."

"Don't giggle me."

"I said you No."

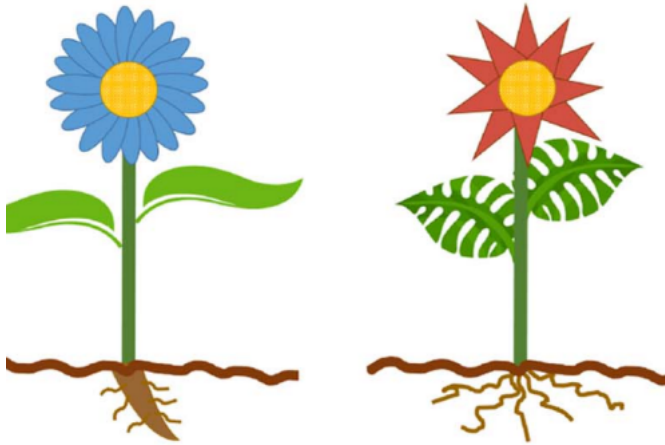
"They delivered you a lot of pizzas."

Simon (1955) again

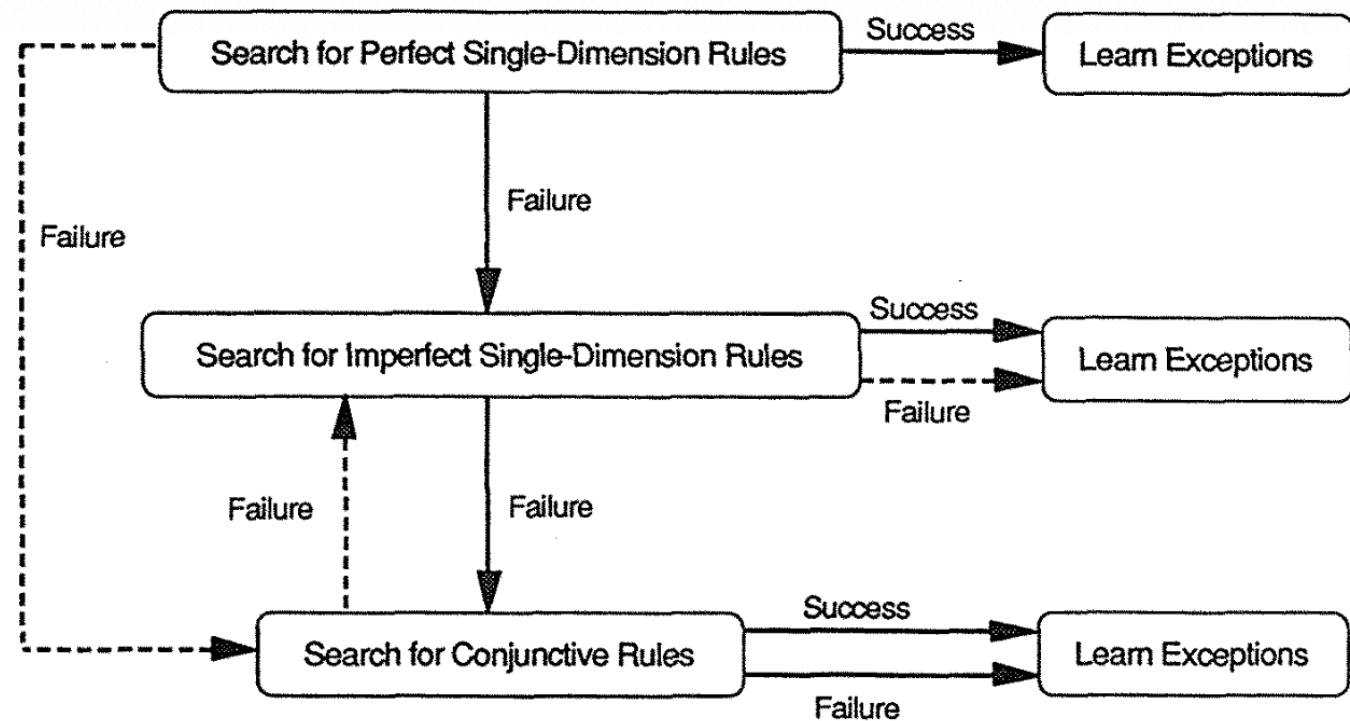
Broadly stated, the task is to replace the global rationality of economic man with a kind of rational behavior that is compatible with the access to information and the computational capacities that are actually possessed by organisms, including man, in the kinds of environments in which such organisms exist. One is tempted to turn

to the literature of psychology for the answer. Psychologists have certainly been concerned with rational behavior, particularly in their interest in learning phenomena. But the distance is so great between our present psychological knowledge of the learning and choice processes and the kinds of knowledge needed for economic and administrative theory that a marking stone placed halfway between might help travellers from both directions to keep to their courses.

Rules and Exceptions



Izidorczyk & Bröder (2022)



Nosofsky et al. (1996)

Glossary of **Free Parameters** in RULEX

Parameter	Description
w	Vector of dimension-salience weights (default: all weights set equal to 1.0)
pstor	General storage probability that affects the rate at which rules and exceptions can be added to memory and the probability that individual dimensions are sampled during the exception-formation process
lcrit	Lax criterion for tentatively accepting imperfect simple rules (default: lcrit = .55)
scrit	Strict criterion for forming a permanent, imperfect, single-dimension rule
ccrit	Strict criterion for forming a permanent, imperfect, conjunctive rule (default: ccrit = 1.0)
lwind	Lower test window (default: lwind = number of training items)
uwind	Upper test window (default: uwind = 2 × lwind)
branch	Probability that a subject tests imperfect, single-dimension rules before testing conjunctive rules (default: branch = 1.0)
capac	Capacity-limit parameter that affects the rate at which new exceptions can be added to memory (default: capac = 1.0)
rerr	Response-error parameter giving the probability that a subject makes the opposite response indicated by the extracted rules (default: rerr = 0.0)

Best Fitting Parameters for the RULEX Model Applied to Medin and Schaffer's (1978) Experiments

Data set	Parameters						
	pstor	scrit	ccrit	w ₁	w ₂	w ₃	w ₄
Experiment 3	.55	.55	1.00	.43	.33	.19	.05
Experiment 2	.65	.85	.55	.25	.06	.24	.44
Experiment 4	.58	.78	.55	.50	.25	.13	.13

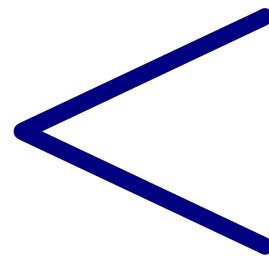
In search for a model

- Most/all models in psychology and cognitive science but with hyper-parameters: not learning models from the data but statistical summaries of the (experimental) data.
- Most/all models in ML/NLP are **optimizing**, in addition to many **hyper-parameters**:
- Many highly frequent rules in language are not learned early, and many infrequent/absent forms in language are used robustly by children.

A faster rule

Rule + Exceptions

- Item 1
- Item 2
- Item 3
- ...
- Item e
- Rule $(N-e)$



Memorize Everything

- Item 1
- Item 2
- Item 3
- ...
- ...
- Item N

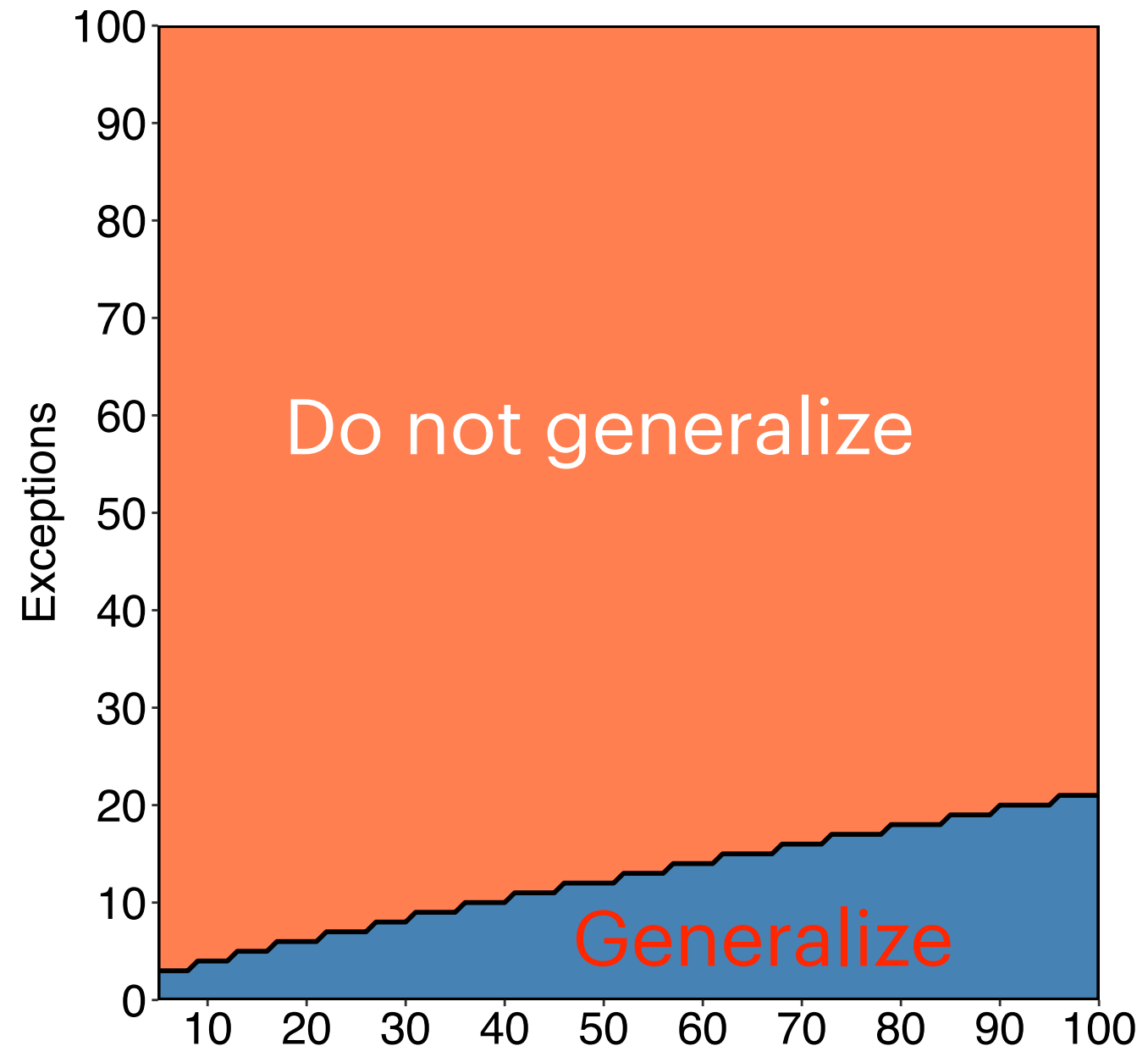
A Truth Ratio

A productive rule applicable to **N** items in the learning data cannot have more than **e** exceptions (Proof by Sam Gutmann)

$$\theta_N = \frac{N}{\ln N}$$

N	θ_N	%
10	4	40%
20	6	30%
50	12	24%
100	21	21%
200	37	19%
500	80	16%
1000	144	14%

Parameter free: no tuning
and runs out-of-the-box



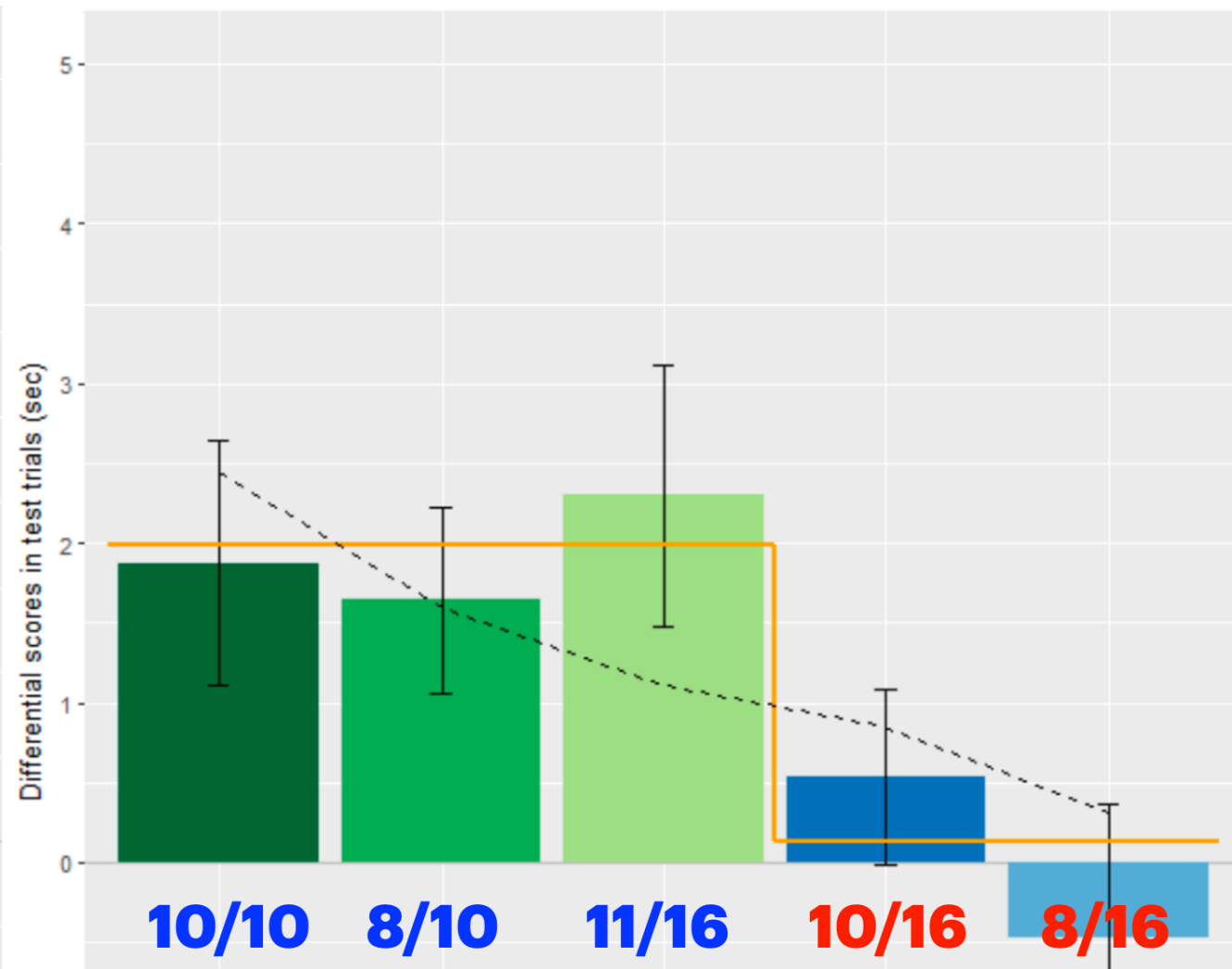
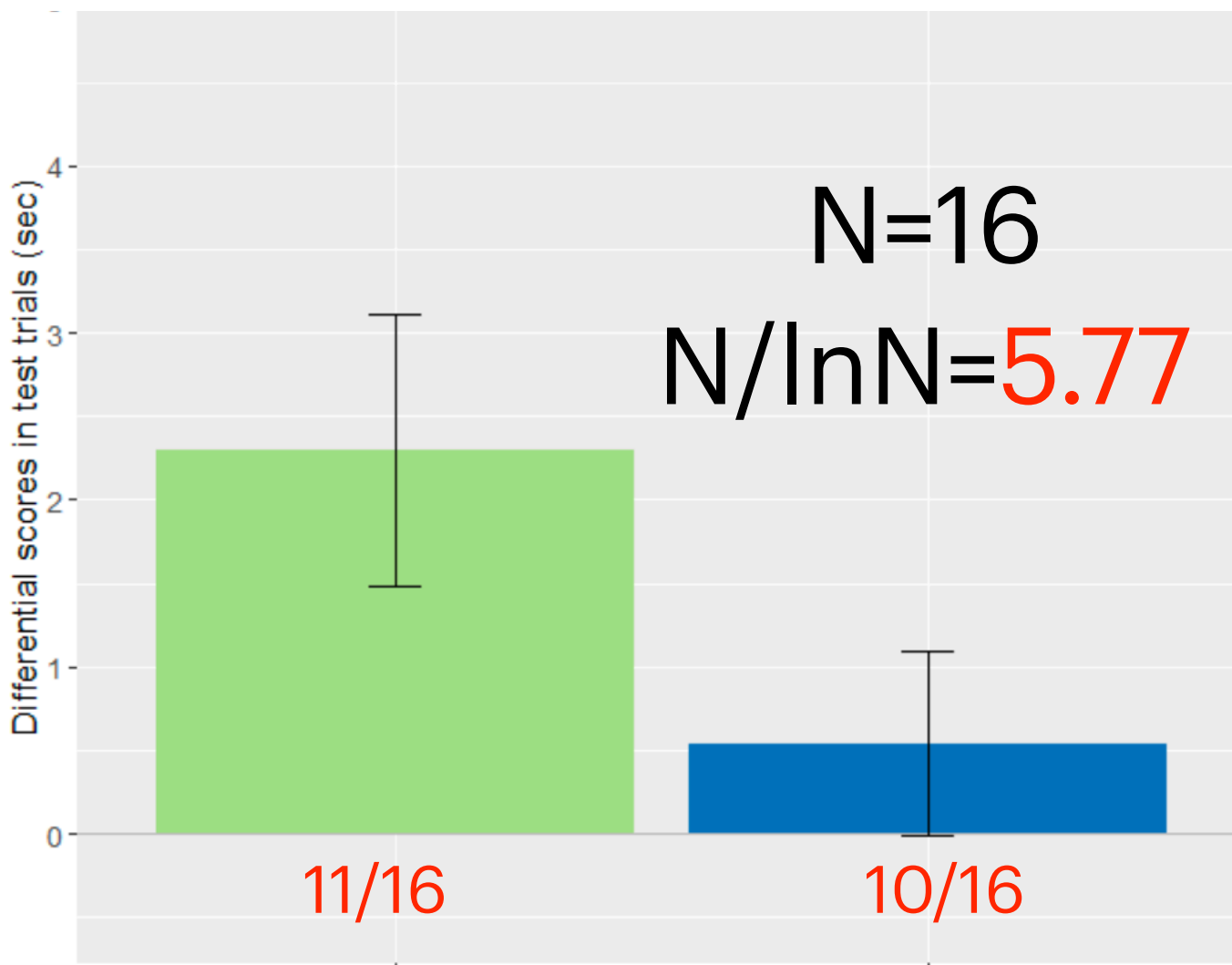
Infants follow the threshold

14-month-old non-Russian learning infants

“Movement” R1: **ABC**→**BAC**; R2: **ABC**→**ACB**

Machty gnutsja lukom → **Gnutsja machty lukom**

Machty gnutsja lukom → **Machty lukom gnutsja**

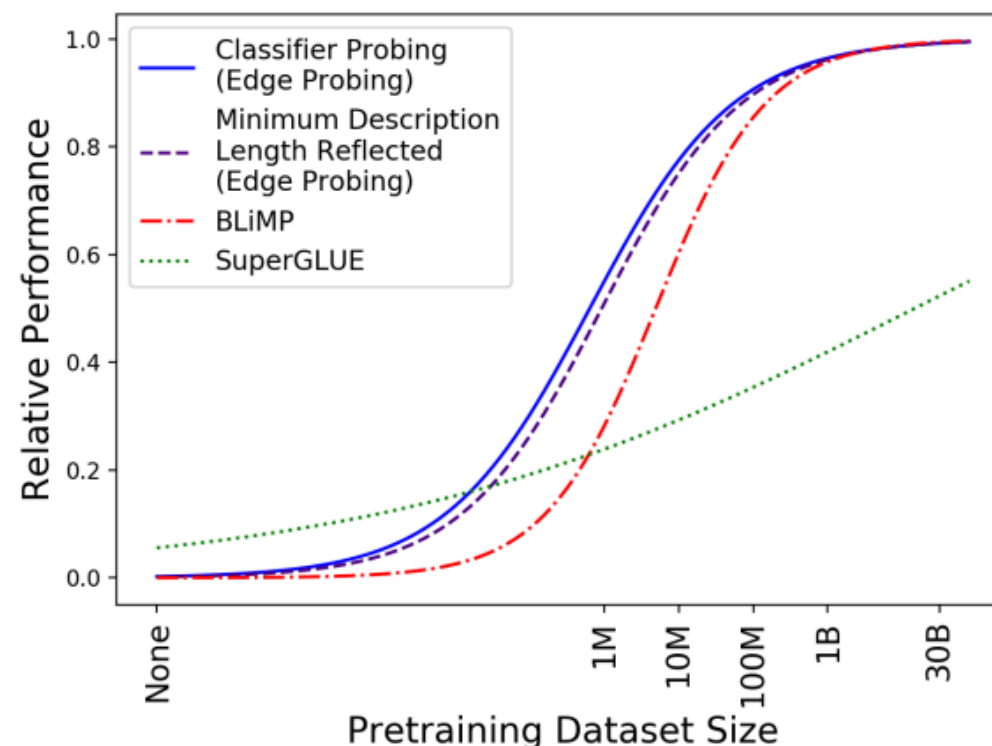


Threshold in Minds

- An unreasonably effective threshold.
- No one has the faintest idea how this is done in the brain or the $N/\ln N$ threshold is computed but the animal world has often turned up surprises (Wittlinger et al. 2006).
- Statistical learning via **transitional probability** (Saffran, Aslin & Newport 1996):
 - $TP(A \rightarrow B) = P(AB)/P(A)$: how well A predicts B
 - Ratio tracking is possible with token frequencies and is of course possible with type frequencies: it may types all along!

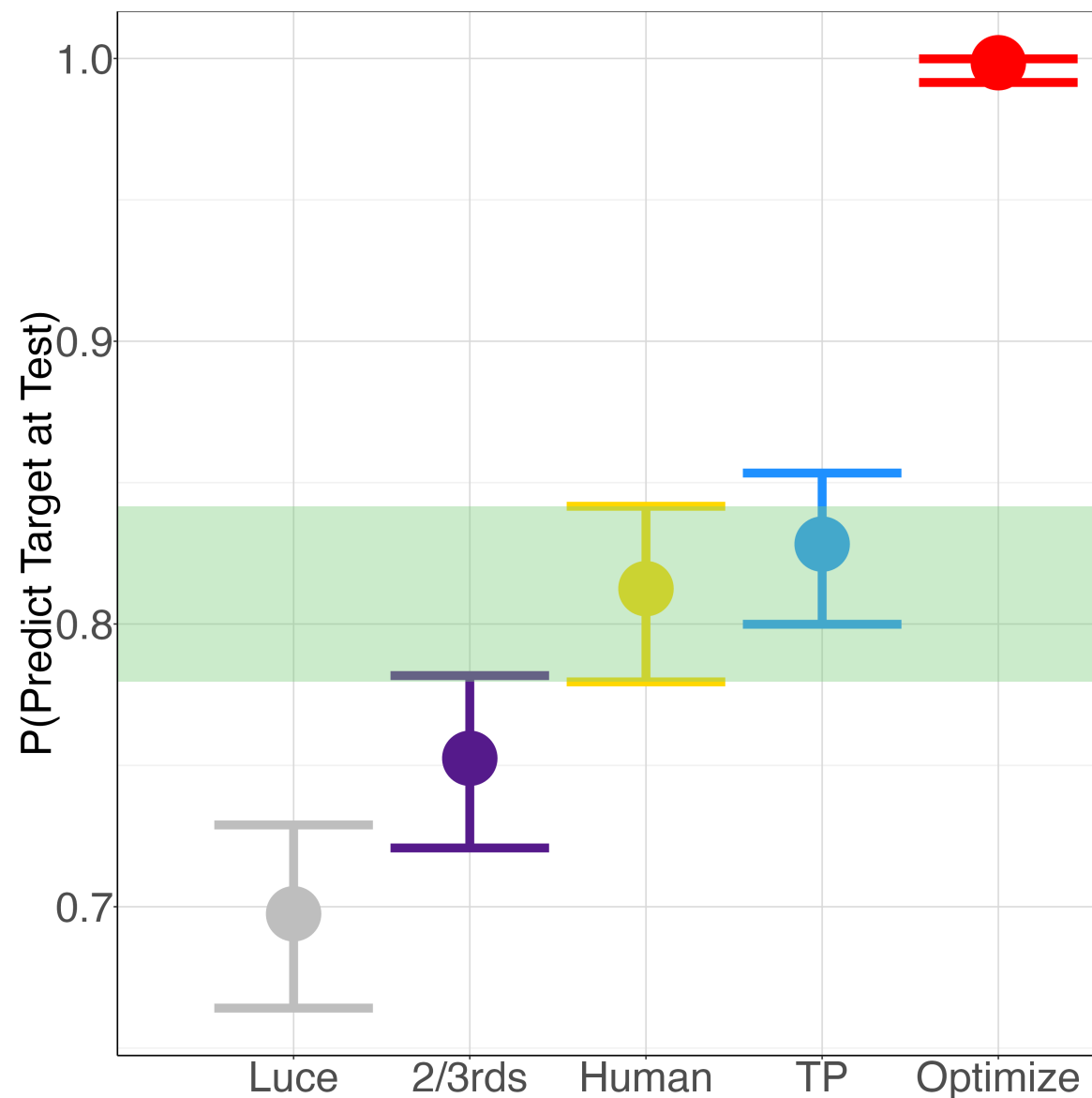
Threshold in machines?

- Friedman, Harnad & Shi (2025): BabyBERTa does **not** show threshold behavior.
- So far as I know, non-monotonic learning curve has never been observed in neural networks (without cheating; Rumelhart & McClelland 1986)



Red and Blue again

A two-stage process



- If the ratio of Reds exceeds the TP threshold where N is the estimated number of items one can hold in memory, **always** select **Red**
- Otherwise **sample** among **Red** and **Blue** proportionally (Luce)

Choice

- A threshold tells us when to stop.
- But we don't always choose, or maximize our choices, despite what they tell us.